

$$Q^a, a=1, \dots, N$$

$$L: P_L Q_\alpha^a = Q_\alpha^a$$

$D=4$: Weyl notation

$$R: P_R \bar{Q}_{a\dot{\alpha}} = (Q_\alpha^a)^*$$

• $SL(2, \mathbb{C})$ indices: $a, \dot{\alpha}$

$$SL(2, \mathbb{C}) \cong SU(2) \otimes \overline{SU(2)}$$

$$\rightarrow \text{Majorana: } Q_D^a = \begin{pmatrix} Q_\alpha^a \\ \bar{Q}^{a\dot{\alpha}} \end{pmatrix} \quad \gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_{\alpha\dot{\beta}} \\ \sigma^\mu^{\dot{\alpha}\beta} & 0 \end{pmatrix}$$

$$\sigma^\mu = (-1, \sigma^i), \quad \bar{\sigma}^\mu = (-1, -\sigma^i) \quad | C^T = -C \quad (D=4)$$

$$(C\gamma^\mu)^T = +C\gamma^\mu \Rightarrow \gamma^{\mu\dagger} C^T = C\gamma^\mu$$

$$(\gamma^\mu)^T = -C\gamma^\mu C^{-1}$$

• Supercharges, for $a=1$, has $U(1)$ symm

$$Q_\alpha \mapsto e^{i\alpha} Q_\alpha, \quad \bar{Q}_{\dot{\alpha}} \mapsto e^{-i\alpha} \bar{Q}_{\dot{\alpha}}$$

$$U(1)_R: [Q_\alpha, R] = q_R Q_\alpha = Q_\alpha \quad | q_R: R\text{-charge}$$

$$[\bar{Q}_{\dot{\alpha}}, R] = \tilde{q}_R \bar{Q}_{\dot{\alpha}} = -\bar{Q}_{\dot{\alpha}} \quad | \tilde{q}_R$$

* Extended susy: $Q_D^a, a=1, \dots, N$

$$Q_\alpha^a \sim \bar{Q}_{b\dot{\beta}}$$

$$\{Q_\alpha^a, \bar{Q}_{b\dot{\beta}}\} = 2\sigma^\mu_{\alpha\dot{\beta}} P_\mu \delta^a_b$$

$$\{Q_\alpha^a, Q_\beta^b\} = \epsilon_{\alpha\beta} Z^{ab}$$

$$\{\bar{Q}_{a\dot{\alpha}}, \bar{Q}_{b\dot{\beta}}\} = \epsilon_{\dot{\alpha}\dot{\beta}} \bar{Z}_{ab}$$

$$Z^{ab} = -Z^{ba}$$

$$Z^{ab} = (\bar{Z}^T)_{ab}$$

$$\text{since } \bar{Q}_{a\dot{\alpha}} = (Q_\alpha^a)^*$$

* Has $U(N)$ symmetry: $R \in U(N), (R)^\dagger_b$

$$Q_\alpha^a \mapsto Q'^a_\alpha = R^a_b Q_\alpha^b$$

$$\bar{Q}_{a\dot{\alpha}} \mapsto \bar{Q}'_{a\dot{\alpha}} = \bar{Q}_{b\dot{\alpha}} (R^\dagger)^b_a$$

a, b : $U(N)$ indices and so Q_α^a & $\bar{Q}_{b\dot{\beta}}$ transform in the fundamental rep $\underline{N}$ and conj rep $\bar{\underline{N}}$ of $U(N)$ respectively.

R -symmetry rotates each Q into each other,

$$(R)^\dagger_b{}^a = (e^{i\lambda^j T_j})^a_b$$

$$[Q_\alpha, R] = [Q_\alpha, 1] + i[Q_\alpha, \lambda^j T_j] = Q_\alpha$$

$$[Q_\alpha^a, T^j] = (B^j)^a_b Q_\alpha^b$$

$$[T^i, T^j] = i f^{ijk} T^k$$

$$[\bar{Q}_{a\dot{\alpha}}, T^j] = -B^j_a{}^b \bar{Q}_{b\dot{\alpha}}$$

* $Z^{ab} \in N \times N \Rightarrow$

$$Z^{ab} = A^{ab}_j T^j$$

$$Z^{ab} \in U(N) \quad | \quad B^{ia}_b A^{bc}_j = -A^{ab}_j B^{ic}_b$$

* If $Z^{ab} = 0$, the maximal R-sym is $U(N)$

$Z^{ab} \neq 0$, then maximal R-sym is $USp(N)$

Reps:

$$P^2 = P_\mu P^\mu, \quad W = W_\mu W^\mu \text{ (Pauli-Lubanski vector)}$$

$$\rightarrow W^2 = W_\mu W^\mu, \quad W^2 = C_{\mu\nu} C^{\mu\nu}$$

$$C_{\mu\nu} = W_\mu P_\nu - W_\nu P_\mu$$

$$\rightarrow \text{For susy} \quad \tilde{W}_\mu = W_\mu - \frac{1}{4} \bar{Q}_{a\dot{\alpha}} \sigma_\mu^{\dot{\alpha}\alpha} Q^a_\alpha$$

$$\tilde{C}_{\mu\nu} = \tilde{W}_\mu P_\nu - \tilde{W}_\nu P_\mu$$

$$\text{then} \quad W^2 = \tilde{C}_{\mu\nu} \tilde{C}^{\mu\nu}$$

* Note: The mass of all fields in a susy-multiplet are the same since $P^\mu P_\mu$ is a Casimir op. of the susy

$$[Q_\alpha, P^\mu] = 0 \Rightarrow [Q, P^2] = 0$$

$$\text{ex: } \left. \begin{aligned} P^2(Q|B\rangle) &= P^2|F\rangle = -m_F^2|F\rangle \\ L = Q P^2|B\rangle &= -m_B^2 Q|B\rangle = -m_B^2|F\rangle \end{aligned} \right\} \underline{m_B = m_F}$$

• For a gauge theory: Let G be a gauge group w/ generators T_G , then

$$[Q_\alpha, T_G] = 0 = [\bar{Q}_{\dot{\alpha}}, T_G]$$

and so all fields in the same susy-multiplet are in the same rep of the Gauge group.

• Bosonic D.O.F. = Ferm. D.O.F.

$$\rightarrow V \equiv (-1)^F \text{ (fermion number operator)}$$

$$V|B\rangle = |B\rangle$$

$$V|F\rangle = -|F\rangle$$

Note: $Q(V|B\rangle) = |F\rangle$

$$V(Q|B\rangle) = V(|F\rangle) = -|F\rangle$$

$$[Q, V]|B\rangle = 2|F\rangle = 2Q|B\rangle$$

$$[Q, V] = 2Q$$

$$\text{Tr}(V) = \sum_B \langle B|V|B\rangle + \sum_F \langle F|V|F\rangle$$

$$= \sum_B \langle B|B \rangle - \sum_F \langle F|F \rangle$$

$$= n_B - n_F = 0 \Rightarrow n_B = n_F$$

Massless Keps: $|p^\mu, \lambda\rangle$, p^μ : momentum, λ : helicity

$$p^\mu = (E, 0, 0, E), \quad \lambda \text{ is an eigenvalue of } J_{12} \quad | \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$J_{12} |p^\mu, \lambda\rangle = \lambda |p^\mu, \lambda\rangle, \text{ then } p^\mu p_\mu = 0$$

$$\{Q_\alpha^a, \bar{Q}_{\dot{\alpha}i}\} = 2\delta_b^a \sigma_{\alpha\dot{\alpha}}^\mu P_\mu = 2\delta_b^a (\sigma^0_{\alpha\dot{\alpha}}(E) + \sigma^4_{\alpha\dot{\alpha}} P_4) \\ = 2\delta_b^a (+1E + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} E)$$

$$\{Q_\alpha^a, \bar{Q}_{\dot{\alpha}i}\} = 4\delta_b^a E \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}_{\alpha\dot{\alpha}}$$

$\therefore$

$$\{Q_1^a, \bar{Q}_{\dot{\alpha}i}\} = 4\delta_b^a E$$

$$\{Q_2^a, \bar{Q}_{\dot{\alpha}i}\} = 0$$

$$\{Q_2^a, \bar{Q}_{\dot{\alpha}i}\} = 0$$

$$\{Q_2^a, \bar{Q}_{\dot{\alpha}i}\} = 0$$

$$\Rightarrow \langle p^\mu, \lambda | \{Q_2^a, \bar{Q}_{\dot{\alpha}i}\} | \tilde{p}^\mu, \tilde{\lambda} \rangle = \langle p^\mu, \lambda | Q_2^a \bar{Q}_{\dot{\alpha}i} | \tilde{p}^\mu, \tilde{\lambda} \rangle + \langle p^\mu, \lambda | \bar{Q}_{\dot{\alpha}i} Q_2^a | \tilde{p}^\mu, \tilde{\lambda} \rangle$$

$$\text{Thus, } Q_2^a |0\rangle = 0 \text{ since } Q_2^a |p^\mu, \lambda\rangle = Q p^\mu |0\rangle = p^\mu Q |0\rangle$$

$$\bullet \{Q_1^a, Q_{\dot{\alpha}i}\} = \delta_b^a 4E \Rightarrow a^b = \frac{Q_1^b}{2\sqrt{E}}, \quad a_b^\dagger = \frac{\bar{Q}_{\dot{\alpha}i}}{2\sqrt{E}}$$

$$\{a^c, a_b^\dagger\} = \delta_c^b, \quad \{a^c, a^b\} = \{a_b^\dagger, a_c^\dagger\} = 0$$

$$\sigma_{\alpha\dot{\alpha}}^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\text{and so, } [Q_\alpha^a, J^{\mu\nu}] = (\sigma^{\mu\nu})_{\alpha}^{\beta} Q_\beta^a$$

$$[Q_\alpha^a, J_{12}] = (\sigma_{12})_{\alpha}^{\beta} Q_\beta^a = \frac{1}{2} (\sigma_3)_{\alpha}^{\beta} Q_\beta^a$$

$$\Rightarrow [Q_1^a, J_{12}] = \frac{1}{2} (\sigma_3)_1^2 Q_2^a$$

$$[Q_1^a, J_{12}] |p^\mu, \lambda\rangle = (Q_1^a J_{12} - J_{12} Q_1^a) |p^\mu, \lambda\rangle$$

$$= (\lambda - J_{12}) (Q_1^a |p^\mu, \lambda\rangle) = \frac{1}{2} Q_1^a |p^\mu, \lambda\rangle$$

$$\Rightarrow J_{12} (Q_1^a |p^\mu, \lambda\rangle) = (\lambda - \frac{1}{2}) (Q_1^a |p^\mu, \lambda\rangle)$$

$$\underline{Q_1^a / a^b}: \text{lowers the helicity by } (\lambda - \frac{1}{2})$$

$$\Rightarrow \underline{\bar{Q}_{\dot{\alpha}i} / a_b^\dagger}: \text{raises the helicity by } (\lambda + \frac{1}{2})$$

* Multiplets: has a minimum helicity state (lowest weight state)

$$|\Omega\rangle$$

$$\underline{N=1}: Q_1 |\Omega\rangle = 0$$

$$a^\dagger |\Omega\rangle = |p^\mu, \lambda + \frac{1}{2}\rangle$$

but $\{\bar{Q}, \bar{Q}\} = 0$ and so $(q^\dagger)^2 |\Omega\rangle = 0$

• so we only have two states

$$|\Omega\rangle = |p^\mu, \lambda\rangle, \quad a^\dagger |\Omega\rangle = |p^\mu, \lambda + \frac{1}{2}\rangle$$

* A QFT requires CPT invariance

CPT-complete $N=1$ multiplet

$$\begin{cases} |\Omega\rangle = |p^\mu, \lambda\rangle \\ a^\dagger |\Omega\rangle = |p^\mu, \lambda + \frac{1}{2}\rangle \\ |-\Omega\rangle = |p^\mu, -\lambda\rangle \\ a |-\Omega\rangle = |p^\mu, -\lambda - \frac{1}{2}\rangle \end{cases}$$

(Bosonic, Fermionic)

($N=1$)

$\lambda=0$: Chiral Multiplet:

$\lambda = 1/2$ 1 $a^\dagger |0\rangle$

$\lambda = 0$ 2 $(1+1) |0\rangle$

$\lambda = -1/2$ 1 $| -1/2 \rangle$ ($N=1$)

How do I get here with the algebra? (cant)

$\lambda = 1/2$ Vector Multiplet:

$\lambda = 1$ $a^\dagger |\Omega\rangle = |p^\mu, 1\rangle$

$\lambda = 1/2$ $|\Omega\rangle = |p^\mu, 1/2\rangle$

—

$\lambda = -1/2$ $|-\Omega\rangle = |p^\mu, -1/2\rangle$

$\lambda = -1$ $a |-\Omega\rangle = |p^\mu, -1\rangle$

* $|\lambda| \leq 1$: Gauge Multiplets, lets look at $N=2$

$N=2, \lambda=0$ (Gauge + CPT)

$|\Omega\rangle = |p^\mu, 0\rangle$

$\lambda=1$ $a_1^\dagger a_2^\dagger |\Omega\rangle$ 1

$\lambda=1/2$ $a_1^\dagger |\Omega\rangle, a_2^\dagger |\Omega\rangle$ 1+1

$\lambda=0$ $|\Omega\rangle, |-\Omega\rangle$ 2 4 + 4

$\lambda=-1/2$ $a_1 |-\Omega\rangle, a_2 |-\Omega\rangle$ 1+1 B F

$\lambda=-1$ $a_1, a_2 |-\Omega\rangle$ 1

$N=2, \lambda=1/2$:

$\lambda=3/2$ $a_1^\dagger a_2^\dagger |\Omega\rangle$

$\lambda=1$ $a_1^\dagger |\Omega\rangle, a_2^\dagger |\Omega\rangle$ 4 + 4

$$\lambda = 1/2 \quad |\Omega\rangle$$

B F

$$\lambda = -1/2 \quad |-\Omega\rangle$$

$$\lambda = -1$$

$$\lambda = -3/2$$

$$2 \cdot 2^N$$

CPT, General

$$N=2, \quad \lambda = -1/2 :$$

$$\underline{4} + \underline{4}$$

B F

$$\begin{cases} \lambda = 1/2 & a_1^\dagger a_2^\dagger |\Omega\rangle + a_1^\dagger a_2^\dagger |-\Omega\rangle \\ \lambda = 0 & a_1^\dagger |\Omega\rangle, a_2^\dagger |\Omega\rangle, a_1^\dagger |-\Omega\rangle, a_2^\dagger |-\Omega\rangle \\ \lambda = -1/2 & |\Omega\rangle, |-\Omega\rangle \end{cases}$$

$$N=3: \quad 2 \cdot 2^3 = \underline{8} + \underline{8}$$

$$N=4: \quad \underline{16} + \underline{16}$$

ex! $N=4$

	$s=2$	$s=3/2$	$s=1$	$s=1/2$	$s=0$
$s_{\max} = 2$	1	4	6	4	1+1
$s_{\max} = 3/2$		1	4	6+1	4+4
$s_{\max} = 1$			1	4	6

For a lowest weight (spin) state $|\Omega\rangle$

$$|\Omega\rangle: \quad \binom{N}{0} = 1$$

$$\lambda_0$$

$$a^\dagger |\Omega\rangle: \quad \binom{N}{1} = N$$

$$\lambda + \frac{1}{2}$$

$$a^\dagger a^\dagger |\Omega\rangle: \quad \binom{N}{2} = \frac{1}{2} N(N-1)$$

⋮

$$a^\dagger a^\dagger \dots a^\dagger |\Omega\rangle: \quad \binom{N}{N} = 1$$

$$\lambda_0 + \frac{N}{2}$$

ex: $N=4$:

$$s_{\max} = 2 = \lambda_0 + \frac{4}{2} = \lambda_0 + 2 \Rightarrow \lambda_0 = 0$$

$$\begin{aligned} s_{\max} &= 1/2 = \lambda_0 + 2 \Rightarrow \lambda_0 = -1/2 \\ s_{\max} &= 1 = \lambda_0 + 2 \Rightarrow \lambda_0 = -1 \end{aligned}$$

$N=2$: (Hypermultiplet) $s_{\max} = 1/2 = \lambda_0 + 1 \Rightarrow \lambda_0 = -1/2$
 [as above]

• so for $N=4$ if we want a theory with gravity we can take the gravity multiplet above.

• For a Pure-Gauge $N=4$ we take the state with $s_{\max}=1$

states: $2 \cdot 2^N = 2 \cdot 2^4 = 16 + 16 = 32$

which is exactly the # of superstates for $N=1$ in $D=10$

Type IIA & IIB since $\alpha = 1, \dots, 8$, $\tilde{\alpha} = 1, \dots, 8$

$$\dim([0010]_2) = 8 = \dim([0001]_8)$$

Massive Reps with Vanishing Central Charge: $p^\mu = (M, 0, 0, 0)$

$$|p^\mu, s, s_3\rangle, Z=0$$

$$\begin{aligned} \{Q_\alpha^a, \bar{Q}_{b\tilde{\alpha}}\} &= 2(\sigma^\mu)_{\alpha\tilde{\alpha}} P_\mu \delta_b^a = 2(-1)(\sigma^0)_{\alpha\tilde{\alpha}} M \delta_b^a \\ &= 2m \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}_{\alpha\tilde{\alpha}} \delta_b^a \end{aligned}$$

$$\{Q_1^a, \bar{Q}_{1b}\} = 2m \delta_b^a$$

$$\{Q_2^a, \bar{Q}_{2b}\} = 2m \delta_b^a$$

$$\{Q_1^a, \bar{Q}_{2b}\} = 0$$

$$\{Q_2^a, \bar{Q}_{1b}\} = 0$$

→ (2-sets of Operators)

→ Cannot conclude $Q_2^a |p^\mu, s, s_3\rangle = 0$ (in general) for all $a=1, \dots, N$

(and so naturally we have more states)

→ Can be easily seen since they're irreps of $SO(D-1)$

and the massless reps were irreps of $SO(D-2)$

$$a_\alpha^b \equiv \frac{Q_\alpha^b}{\sqrt{2m}}, \quad (a^\dagger)_{\alpha\tilde{\alpha}} = \frac{\bar{Q}_{\alpha\tilde{\alpha}}}{\sqrt{2m}}$$

$a_\alpha^b |Q\rangle = 0$ and for $\tilde{\alpha} \in \{1, 2\}$ we have $2N$ creation ops so we have 2^{2N} states

$$\left(\begin{array}{c} 2^{2N} : \text{Massive} \\ (2 \text{ sets of } Q_i) \end{array} \right)$$

$$\left(\begin{array}{c} 2^N : \text{Massless} \\ (1 \text{ set } Q_i) \end{array} \right)$$

Massive Reps with non-vanishing Z :

- For Z , the massive reps are shortened: branes break translational symmetry so we get goldstone modes

$\rightarrow Z$ has eigenvalues q^i and made into an $N \times N$ -matrix

$$\underline{N=2}: Z^{ab} = \begin{pmatrix} 0 & q_1 \\ -q_1 & 0 \end{pmatrix}$$

$$\underline{N>2}: Z^{ab} = \begin{pmatrix} 0 & q_1 & 0 & \dots & 0 \\ -q_1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & q_2 & \dots \\ 0 & 0 & -q_2 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \\ 0 & \dots & 0 & q_{N/2} & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix}$$

q_a : central charges

* Even N : $\tilde{Q}_{\alpha\pm}^i \equiv (Q_{\alpha}^{2i-1} \pm (Q_{\alpha}^{2i})^\dagger)$ $j = 1, \dots, N/2$

$$\begin{aligned} \{\tilde{Q}_{\alpha+}^i, (\tilde{Q}_{\beta+}^j)\} &= \delta_j^i \delta_{\beta\alpha} (2m + q_j) \\ \{\tilde{Q}_{\alpha-}^i, \tilde{Q}_{\beta-}^j\} &= \delta_j^i \delta_{\alpha\beta} (2m - q_j) \end{aligned}$$

• For unitary reps $(2m \pm q_j) \geq 0 \Rightarrow$

BPS - Bound: $|q_j| = 2m$

$$|q_j| \leq 2m$$

$j = 1, \dots, N/2$

* Suppose that of the $N/2$ terms K are BPS

$$K \begin{cases} q_i = \pm 2m \\ \vdots \\ q_{i'} = \pm 2m \end{cases}$$

then we are left with $(2N - 2K)$ creation ops and thus $2^{2(N-K)}$ states

$\Rightarrow \frac{1}{2}$ BPS multiplets

BPS - Multiplets

$K=0$	2^{2N} (states)	<u>long</u> multiplet
$0 < K < N/2$	$2^{2(N-K)}$ (states)	<u>short</u> multiplet
$K = N/2$	2^N (states)	<u>ultrashort</u> multiplet

* For Black Holes: BPS states correspond to extremal B.H.s where $M=Q$ and solutions of extended sugra

$N=1$ Super space Formalism: $\mathbb{R}^4 \rightarrow \mathbb{R}^{4|4}$, $Z^A = (x^m, \theta_\alpha, \bar{\theta}_{\dot{\alpha}})$

• (super-susy) ops: $G(x, \theta, \bar{\theta}) \equiv \exp(-ix^m P_m + i\theta Q + i\bar{\theta} \bar{Q})$

with $\theta^\alpha Q_\alpha$ and $\bar{\theta}_{\dot{\alpha}} \bar{Q}^{\dot{\alpha}}$

$$* g(\xi, \bar{\xi}) = G(\omega, \xi, \bar{\xi}) = e^{i\xi Q + i\bar{\xi} \bar{Q}}$$

and so $g(\xi, \bar{\xi}) G(x, \theta, \bar{\theta}) = G(x^\mu + i\theta\sigma^\mu_{\alpha\dot{\beta}}\xi - i\bar{\xi}\sigma^\mu_{\alpha\dot{\beta}}\bar{\theta}, \theta + \xi, \bar{\theta} + \bar{\xi})$

$$g(\xi, \bar{\xi}) : (x^\mu, \theta^\alpha, \bar{\theta}_{\dot{\alpha}}) \mapsto (x^\mu + i\theta\sigma^\mu_{\alpha\dot{\beta}}\xi - i\bar{\xi}\sigma^\mu_{\alpha\dot{\beta}}\bar{\theta}, \theta + \xi, \bar{\theta} + \bar{\xi})$$

$$\delta x^\mu = \xi^\alpha Q_\alpha(x^\mu) + \bar{\xi}_{\dot{\alpha}} \bar{Q}^{\dot{\alpha}}(x^\mu)$$

$$Q_\alpha(x^\mu) = -i\sigma^\mu_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}} \Rightarrow Q_\alpha(x^\mu) = -i\sigma^\mu_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}}\partial_\nu(x^\mu)$$

$$\bar{Q}_{\dot{\alpha}}(x^\mu) = -i\theta^\alpha\sigma^\mu_{\alpha\dot{\beta}} \Rightarrow \bar{Q}_{\dot{\alpha}}(x^\mu) = -i\theta^\alpha\sigma^\mu_{\alpha\dot{\beta}}\partial_\nu(x^\mu)$$

$$\delta\theta^\alpha = \xi^\beta Q_\beta(\theta^\alpha) + \bar{\xi}_{\dot{\beta}} \bar{Q}^{\dot{\beta}}(\theta^\alpha)$$

$$\Rightarrow Q_\beta(\theta^\alpha) = \delta^\alpha_\beta = \frac{\partial}{\partial\theta^\beta}\theta^\alpha$$

$$\Rightarrow \boxed{\begin{aligned} Q_\alpha &= \frac{\partial}{\partial\theta^\alpha} - i\sigma^\mu_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}}\partial_\mu \\ \bar{Q}_{\dot{\alpha}} &= \frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} - i\theta^\alpha\sigma^\mu_{\alpha\dot{\beta}}\epsilon^{\dot{\beta}\alpha}\partial_\mu \end{aligned}}$$

* Consider $G(x, \theta, \bar{\theta})$ from the right $G(x, \theta, \bar{\theta}) g(\xi, \bar{\xi})$, then

$$G(x, \theta, \bar{\theta}) g(\xi, \bar{\xi}) : \xi + \bar{\xi} + () + \frac{1}{2} [X, Y] + \frac{1}{12} \dots$$

$$[-ix^\mu P_\mu + i\theta Q + i\bar{\theta} \bar{Q}, i\xi Q + i\bar{\xi} \bar{Q}]$$

$$= (x^\mu - i\theta\sigma^\mu_{\alpha\dot{\beta}}\xi + i\bar{\xi}\sigma^\mu_{\alpha\dot{\beta}}\bar{\theta}, \theta + \xi, \bar{\theta} + \bar{\xi})$$

$$\delta = \xi D + \bar{\xi} \bar{D}$$

$$D_\alpha = \frac{\partial}{\partial\theta^\alpha} + i\sigma^\mu_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}}\partial_\mu$$

$$\bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} - i\theta^\alpha\sigma^\mu_{\alpha\dot{\beta}}\epsilon^{\dot{\beta}\alpha}\partial_\mu$$

$$\{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} = 2i\sigma^\mu_{\alpha\dot{\beta}}\partial_\mu$$

$$\{D_\alpha, \bar{D}_{\dot{\alpha}}\} = -2i\sigma^\mu_{\alpha\dot{\beta}}\partial_\mu$$

• all others zero.

• General scalar field: $F(x, \theta, \bar{\theta}) = \psi(x) + \theta\psi(x) + \bar{\theta}\bar{\psi}(x) + \theta\theta M(x) + \bar{\theta}\bar{\theta} N(x) + (\theta\sigma^\mu\bar{\theta})V_\mu(x) + \theta\theta(\bar{\theta}\bar{\lambda}) + (\bar{\theta}\bar{\theta})\theta\rho(x) + (\theta\theta)(\bar{\theta}\bar{\theta})D(x)$

D-term, F-terms

$$\text{Susy: } \delta_\epsilon F = \delta_\epsilon \psi + \theta \delta_\epsilon \psi + \dots + \theta^2 \bar{\theta}^2 \delta_\epsilon D(x) = (\epsilon Q + \bar{\epsilon} \bar{Q}) F$$

using Q_α we find the susy transformations.

Types of superfields:

(i) chiral

$\bar{D}_{\dot{\alpha}} \Phi = 0$

(ii) Antichiral

$$D_{\alpha} \bar{\Phi} = 0$$

(iii) Vector / Real

$$V = V^{\dagger}$$

(iv) Linear

$$DDL = 0 \quad \oplus \quad L = L^{\dagger}$$

Chiral Superfield: $\bar{D}_{\dot{\alpha}} \Phi = 0$ $s_{\max} = 1/2$

• Define mixed coords: $y_{\pm}^m = x^m \pm i\theta\sigma^m\bar{\theta}$

$$\bar{D}_{\dot{\alpha}} \Phi = (\bar{D}_{\dot{\alpha}} \bar{\theta}^{\dot{\beta}}) \frac{\partial \Phi}{\partial \bar{\theta}^{\dot{\beta}}} \Big|_{y, \bar{\theta}} + (\bar{D}_{\dot{\alpha}} y^m) \frac{\partial \Phi}{\partial y^m} \Big|_{\theta, \bar{\theta}} + (\bar{D}_{\dot{\alpha}} \bar{\theta}^{\dot{\beta}}) \frac{\partial \Phi}{\partial \bar{\theta}^{\dot{\beta}}} \Big|_{y, \theta}$$

$$\bullet D_{\alpha} = -\frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} - i\theta^{\beta} \sigma_{\alpha\dot{\beta}}^m \partial_m$$

$$\text{so } D_{\alpha} \bar{\theta}^{\dot{\alpha}} = 0$$

$$\begin{aligned} \bar{D}_{\dot{\alpha}} y^m &= \left(-\frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} - i\theta^{\beta} \sigma_{\alpha\dot{\beta}}^m \partial_{\beta} \right) (x^m + i\theta\sigma^m\bar{\theta}) \\ &= -i\theta^{\alpha} \sigma_{\alpha\dot{\alpha}}^m + i\theta^{\alpha} \sigma_{\alpha\dot{\alpha}}^m = 0 \end{aligned}$$

$$\bar{D}_{\dot{\alpha}} \bar{\theta}^{\dot{\beta}} = -\delta_{\dot{\alpha}}^{\dot{\beta}} \left(\frac{\partial \Phi}{\partial \bar{\theta}^{\dot{\beta}}} \right) = 0$$

Thus, the chiral condition implies:

$$\frac{\partial \Phi}{\partial \bar{\theta}^{\dot{\beta}}} = 0$$

$$\bar{\Phi}(y, \theta) = \varphi(y) + \bar{\theta} \psi(y) + \bar{\theta}^2 F(y)$$

$$\bullet D_{\alpha} = \frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} + 2i\sigma_{\alpha\dot{\alpha}}^m \bar{\theta}^{\dot{\beta}} \frac{\partial}{\partial y^m}, \quad \bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}}$$

Anti-chiral Multiplet: $\Phi^{\dagger}$

$$\Phi^{\dagger} = \Phi^{\dagger} \quad \text{and} \quad D^{\dagger} = \bar{D}$$

Vector Superfield: $V(x, \theta, \bar{\theta}) = V^{\dagger}(x, \theta, \bar{\theta})$

$$\left(f^1(x) + \theta f^2(x) + \bar{\theta} \bar{f}^3(x) + \theta^2 f^4(x) + \bar{\theta}^2 \bar{f}^5(x) + \theta\sigma^m\bar{\theta} f_m^{(6)}(x) + \theta^2 \bar{\theta} \bar{f}^{(7)}(x) + \bar{\theta}^2 \theta f^{(8)}(x) + \theta^2 \bar{\theta}^2 f^{(9)}(x) \right)^{\dagger}$$

$$\bullet \underline{f^1(x) = \bar{f}^{(1)}(x)} \quad (\text{Real})$$

$$\bullet \bar{\theta} \bar{f}^{(2)} = \bar{\theta} \bar{f}^{(3)} \Rightarrow \underline{f^{(2)} = f^{(3)}}$$

$$\bullet \bar{\theta}^2 \bar{f}^{(4)} = \bar{\theta}^2 \bar{f}^{(5)} \Rightarrow \underline{\bar{f}^{(4)} = \bar{f}^{(5)}}$$

$$\bullet \theta\sigma^m\bar{\theta} \bar{f}_m^{(6)}(x) = \theta\sigma^m\bar{\theta} f_m^{(6)}(x) \Rightarrow \underline{\bar{f}^{(6)} = f^{(6)}}$$

$$\bullet \bar{f}^{(8)} = f^{(7)}$$

$$\underline{\bar{f}^{(9)} = f^{(9)}}$$

$$V(x, \theta, \bar{\theta}) = C(x) + \theta\phi(x) + \bar{\theta}\bar{\chi} + \theta\theta M + \bar{\theta}\bar{\theta} N + \theta\sigma^m\bar{\theta} V_m + (\theta\theta)\bar{\theta}\bar{\theta} \bar{\chi}$$

$$+ (\bar{\theta}\bar{\theta})\theta\psi + \theta^2\bar{\theta}^2 D(x)$$

$$V^*(x, \theta, \bar{\theta}) = C^*(x) + \bar{\theta}\bar{\psi} + \theta\psi + \bar{\theta}^2\bar{M} + \theta^2\bar{N} + \theta\sigma^m\bar{\theta}V_m^* + \bar{\theta}^2\theta\lambda + \theta^2\bar{\theta}\bar{\psi} + \theta^2\bar{\theta}^2 D^*(x)$$

$$(i) \quad C(x) = C^*(x) \quad \Rightarrow \quad C(x) \in \mathbb{R}$$

$$(ic) \quad \phi(x) = \chi(x)$$

$$(iic) \quad M^*(x) = N(x)$$

$$(iv) \quad V_m = V_m^* \quad \Rightarrow \quad V_m(x) \in \mathbb{R}$$

$$(v) \quad \lambda = \psi$$

$$(vi) \quad D(x) = D^*(x) \quad \Rightarrow \quad D(x) \in \mathbb{R}$$

$$V(x, \theta, \bar{\theta}) = C(x) + \theta\phi(x) + \bar{\theta}\bar{\psi} + (\theta^2)M(x) + \bar{\theta}N^* + \theta\sigma^m\bar{\theta}V_m + \theta^2\bar{\theta}\bar{\lambda} + \bar{\theta}^2(\theta\lambda) + \theta^2\bar{\theta}^2 D(x)$$

$$* \text{ Find: } \mathcal{L}_5 D(x) = \frac{i}{2} (\partial_\mu \lambda \sigma^\mu \bar{\epsilon} - \partial_\mu \bar{\lambda}(x) \bar{\sigma}^\mu \epsilon)$$

* Take Gauge fields to be invariant under
 $V \rightarrow V + (\Phi + \Phi^\dagger)$

$$\lambda(x) \rightarrow \lambda(x) - \frac{i}{2} \sigma^\mu \partial_\mu \bar{\phi}(x)$$

$$D(x) \rightarrow D(x) - \frac{1}{4} \square C(x)$$

$$\Phi + \Phi^\dagger = A(x) + A^* + \sqrt{2}\theta\psi + \sqrt{2}\bar{\theta}\bar{\psi} + \dots$$

Further make

$$\left. \begin{aligned} C(x) &\rightarrow C'(x) = C(x) + A(x) + A^*(x) \\ \phi(x) &\rightarrow \phi'(x) = \phi(x) + \sqrt{2}\psi(x) \\ M &\rightarrow M'(x) = M(x) + F(x) \\ V_m &\rightarrow V'_m = V_m + i\partial_\mu (A(x) - A^*(x)) \end{aligned} \right\} \left(\begin{array}{l} \text{U(1) -} \\ \text{Zukino} \\ \text{Gauge} \end{array} \right)$$

$$V_{WZ}(x) = \theta\sigma^\mu\bar{\theta} \left(V_\mu + i\partial_\mu (A(x) - A^*(x)) \right) + \theta^2\bar{\theta}\bar{\lambda} + \bar{\theta}^2\theta\lambda + \theta^2\bar{\theta}^2 D(x)$$

$$\bullet \text{ let } \text{Im}(A) = 0$$

$$V_{WZ} = -\theta\sigma^\mu\bar{\theta}A_\mu + i\theta^2\bar{\theta}\bar{\lambda} - i\bar{\theta}^2\theta\lambda + \frac{1}{2}\theta^2\bar{\theta}^2 D(x)$$

$$V_{WZ} = -\theta\sigma^\mu\bar{\theta}A_\mu + i\theta^2\bar{\theta}\bar{\lambda} - i\bar{\theta}^2\theta\lambda + \frac{1}{2}\theta^2\bar{\theta}^2 D(x)$$

$$V_{W2} = -\theta^\alpha \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\mu - \theta^\alpha \sigma^\nu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\nu$$

$$= \theta^\alpha \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\mu - \theta^\beta \sigma^\nu_{\beta\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} A_\nu$$

$$\theta^\alpha \theta^\beta = -\frac{1}{2} \epsilon^{\alpha\beta} \theta^2$$

$$\bar{\theta}^{\dot{\alpha}} \bar{\theta}^{\dot{\beta}} = \frac{1}{2} \epsilon^{\dot{\alpha}\dot{\beta}} \bar{\theta}^2$$

$$V_{W2} = -\theta^\alpha \theta^\beta (\sigma^\mu_{\alpha\dot{\beta}} \sigma^\nu_{\beta\dot{\alpha}}) \bar{\theta}^{\dot{\beta}} \bar{\theta}^{\dot{\alpha}} A_\mu A_\nu$$

$$= \frac{1}{2} \epsilon^{\alpha\beta} \theta^2 (\sigma^\mu_{\alpha\dot{\beta}} \sigma^\nu_{\beta\dot{\alpha}}) \frac{1}{2} \epsilon^{\dot{\alpha}\dot{\beta}} \bar{\theta}^2 A_\mu A_\nu$$

$$= \frac{1}{4} \theta^2 \bar{\theta}^2 \text{Tr}(\sigma^\mu \sigma^\nu) A_\mu A_\nu$$

$$= \frac{1}{4} \theta^2 \bar{\theta}^2 (-2\eta^{\mu\nu}) A_\mu A_\nu$$

$$= -\frac{1}{2} \theta^2 \bar{\theta}^2 A^\mu A_\mu$$

$$V_{W2} = 0 \quad , \quad n \geq 3$$

Chiral/Anti-chiral Field strengths:

$$W_\alpha \equiv -\frac{1}{4} D^2 D_\alpha V$$

$$\bar{W}_{\dot{\alpha}} \equiv -\frac{1}{4} \bar{D}^2 \bar{D}_{\dot{\alpha}} V$$

* Written using

$$y_+ \equiv x^\mu + i\theta\sigma^\mu\bar{\theta}$$

$$y_- \equiv x^\mu - i\theta\sigma^\mu\bar{\theta}$$

and so

$$D_\alpha = \frac{\partial}{\partial \theta^\alpha} + 2i\sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \frac{\partial}{\partial y_+} \quad , \quad \bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}}$$

$$* e^V = 1 + V + \frac{1}{2} V^2$$

$$= 1 + (\theta\sigma^\mu\bar{\theta})V_\mu + \theta^2 \bar{\theta}^2 + \bar{\theta}^2 \theta^2 + \theta^2 \bar{\theta}^2 \left(D(x) - \frac{1}{4} V_\mu V^\mu \right)$$

* $W_{\alpha\dot{\alpha}}$ is a chiral spinor superfields

Claim: W_α and $\bar{W}_{\dot{\alpha}}$ are L/R-handed chiral superfields and invariant under susy gauge trans.

$$(a) D^3 = \bar{D}^3 = 0$$

Proof: $D_\alpha = \partial_\alpha + i\sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_\mu$

$$D^\alpha D_\alpha = (-\partial^\alpha - i\sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_\mu) (\partial_\alpha + i\sigma^\nu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_\nu)$$

$$= -\partial^\alpha \partial_\alpha - i\sigma^\nu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial^\alpha \partial_\nu - i\sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_\mu \partial_\alpha$$

$$+ \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \sigma^\nu_{\alpha\dot{\gamma}} \bar{\theta}^{\dot{\gamma}} \partial_\mu \partial_\nu$$

$$D_{\dot{\beta}}(\bar{D}^2) = (\partial_{\dot{\beta}} + i\sigma^\mu_{\dot{\beta}\alpha} \theta^\alpha \partial_\mu) (-\partial^{\dot{\alpha}} \partial_{\dot{\alpha}} - \dots + \sigma^\mu \sigma^\nu \theta \partial_\mu \partial_\nu)$$

$$= 0 \dots$$

and likewise for $\bar{D}_{\dot{\alpha}}(\bar{D}_{\dot{\beta}} \bar{D}^{\dot{\beta}}) = 0$

and so $\bar{D}_{\dot{\alpha}} W_{\alpha} = 0$
 $D_{\alpha} \bar{W}_{\dot{\alpha}} = 0$

(ii) Invariant under Gauge transformations

$$\begin{aligned} W_{\alpha} &\rightarrow W'_{\alpha} = -\frac{1}{4}(\bar{D}\bar{D})D_{\alpha}V' = -\frac{1}{4}(\bar{D}\bar{D})D_{\alpha}(V + \Phi + \Phi^{\dagger}) \\ &= -\frac{1}{4}(\bar{D}\bar{D})D_{\alpha}V - \frac{1}{4}\bar{D}\bar{D}D_{\alpha}\Phi \\ &= W_{\alpha} - \frac{1}{4}\bar{D}\bar{D}D_{\alpha}\Phi \\ &= W_{\alpha} + \frac{1}{4}\bar{D}^{\dot{\alpha}}\bar{D}_{\dot{\alpha}}D_{\alpha}\Phi \\ &= W_{\alpha} + \frac{1}{4}\bar{D}^{\dot{\alpha}}(-D_{\alpha}\bar{D}_{\dot{\alpha}} - i\sigma^{\mu}_{\alpha\dot{\alpha}}\partial_{\mu})\Phi \end{aligned}$$

$W'_{\alpha} = W_{\alpha}$ ✓

and likewise $\bar{W}'_{\dot{\alpha}} = \bar{W}_{\dot{\alpha}}$

* Components of W_{α} we use: $y^{\mu} = x^{\mu} + i\theta\sigma^{\mu}\bar{\theta}$ (1)
 $z^{\mu} = x^{\mu} - i\theta\sigma^{\mu}\bar{\theta}$ (2)

$$\begin{aligned} z^{\mu}: &\begin{cases} D_{\alpha}^{(2)} = \partial_{\alpha} \\ \bar{D}_{\dot{\alpha}}^{(2)} = -\partial_{\alpha} - 2i\theta^{\alpha}\sigma^{\mu}_{\alpha\dot{\alpha}}\frac{\partial}{\partial z^{\mu}} \end{cases} \quad \begin{cases} D_{\alpha}z^{\mu} = 0 \\ D_{\alpha}\bar{\theta}_{\dot{\alpha}} = 0 \end{cases} \\ y^{\mu}: &\begin{cases} D_{\alpha}^{(1)} = \partial_{\alpha} + 2i\sigma^{\mu}_{\alpha\dot{\alpha}}\bar{\theta}^{\dot{\alpha}}\frac{\partial}{\partial y^{\mu}} \\ \bar{D}_{\dot{\alpha}}^{(1)} = -\bar{\partial}_{\dot{\alpha}} \end{cases} \quad \begin{cases} \bar{D}_{\dot{\alpha}}y^{\mu} = 0 \\ \bar{D}_{\dot{\alpha}}\theta_{\alpha} = 0 \end{cases} \end{aligned}$$

$V_{WZ}(x, \theta, \bar{\theta}) = (\theta\sigma^{\mu}\bar{\theta})V_{\mu} + \theta^2\bar{\theta}\bar{\lambda}(x) + \bar{\theta}^2\theta\lambda(x) + \theta^2\bar{\theta}^2 D(x)$

$x^{\mu} = y^{\mu} - i\theta\sigma^{\mu}\bar{\theta} = z^{\mu} + i\theta\sigma^{\mu}\bar{\theta}$

$V_{WZ}(y^{\mu} - i\theta\sigma^{\mu}\bar{\theta}, \theta, \bar{\theta}) = \theta\sigma^{\mu}\bar{\theta}V_{\mu}(y - i\theta\sigma\bar{\theta}) + \dots$
 $= \theta\sigma^{\mu}\bar{\theta}[V_{\mu}(y) - (i\theta\sigma^{\lambda}\bar{\theta}\partial_{\lambda})V_{\mu}] + \dots$

all other terms vanish in $\theta\sigma^{\mu}\bar{\theta}$ expansion.

$\rightarrow -i(\theta\sigma^{\mu}\bar{\theta})\theta\sigma^{\lambda}\bar{\theta}\partial_{\lambda}V_{\mu}$
 $= i\theta^{\alpha}\theta^{\beta}\sigma^{\lambda}_{\alpha\dot{\alpha}}\sigma^{\mu}_{\beta\dot{\beta}}\partial_{\lambda}V_{\mu}\bar{\theta}^{\dot{\alpha}}\bar{\theta}^{\dot{\beta}}$
 $= -\frac{1}{4}i\theta^2\bar{\theta}^2\text{Tr}(\sigma^{\lambda}\sigma^{\mu})\partial_{\lambda}V_{\mu}$
 $= -\frac{1}{4}i\theta^2\bar{\theta}^2 2\eta^{\mu\lambda}\partial_{\lambda}V_{\mu}$
 $= \theta^2\bar{\theta}^2[-\frac{1}{2}i\partial_{\mu}V^{\mu}]$

$V_{WZ}^{(1)}(y^{\mu}, \theta, \bar{\theta}) = \theta\sigma^{\mu}\bar{\theta}V_{\mu}(y) + \theta^2\bar{\theta}\bar{\lambda}(y) + \bar{\theta}^2\theta\lambda(y) + \theta^2\bar{\theta}^2[D(y) - \frac{i}{2}\partial_{\mu}V^{\mu}(y)]$ (3)

* z is identical to $y + \frac{1}{2}i\partial_{\mu}V^{\mu}$

$V_{WZ}^{(2)}(z^{\mu}, \theta, \bar{\theta}) = \theta\sigma^{\mu}\bar{\theta}V_{\mu}(z) + \theta^2\bar{\theta}\bar{\lambda}(z) + \bar{\theta}^2\theta\lambda(z) + \theta^2\bar{\theta}^2 D(z)$ (4)

$$V_{wz}(z, \theta, \bar{\theta}) = (\theta \sigma^{\mu} \bar{\theta}) V_{\mu}(z) + \bar{\theta}^2 \theta \lambda(z) + \theta^2 \bar{\theta} \bar{\lambda}(z) + \theta^2 \bar{\theta}^2 \left[D(z) + \frac{i}{2} \partial_{\mu} V^{\mu}(z) \right] \quad (c)$$

* Thus: $W_{\alpha} = -\frac{1}{4} \bar{D}^{(1)} \bar{D}^{(1)} D_{\alpha}^{(1)} V_{wz}^{(1)}(y, \theta, \bar{\theta})$

$$\bullet D_{\alpha}^{(1)} V_{wz}^{(1)}(y, \theta, \bar{\theta}) = \left(\partial_{\alpha} + 2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^{\mu}} \right) \left(V_{wz}^{(1)}(y, \theta, \bar{\theta}) \right) \\ = \left(\partial_{\alpha} + 2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^{\mu}} \right) \left[\theta^{\mu} \bar{\theta} V_{\mu} + \bar{\theta}^2 \theta \lambda + \theta^2 \bar{\theta} \bar{\lambda} + \theta^2 \bar{\theta}^2 \left(D(y) - \frac{i}{2} \partial_{\mu} V^{\mu}(y) \right) \right]$$

$$\textcircled{a} \sigma^{\mu}_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} V_{\mu} + \bar{\theta}^2 \delta_{\alpha}^{\beta} \lambda_{\beta} + \partial_{\alpha} (\theta^2)$$

with

$$\partial_{\alpha} (\theta^{\beta} \theta_{\beta}) = \partial_{\alpha} (\theta^{\beta} \theta^{\gamma} \epsilon_{\gamma\beta}) = \epsilon_{\gamma\beta} (\delta_{\alpha}^{\beta} \theta^{\gamma} - \theta^{\beta} \delta_{\alpha}^{\gamma}) \\ = \epsilon_{\gamma\alpha} \theta^{\gamma} - \theta^{\beta} \epsilon_{\alpha\beta} \\ = 2 \theta_{\alpha}$$

$$\rightarrow \sigma^{\mu}_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} V_{\mu} + \bar{\theta}^2 \lambda_{\alpha} + 2 \theta_{\alpha} \left[\bar{\theta} \bar{\lambda} + \bar{\theta}^2 \left(D(y) - \frac{i}{2} \partial_{\mu} V^{\mu}(y) \right) \right]$$

$$\textcircled{b} 2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \theta^{\beta} \sigma^{\nu}_{\beta\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_{\mu} V_{\nu} + 2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \theta^{\beta} \bar{\theta}^2 \partial_{\mu} \bar{\lambda}$$

$$= \sigma^{\mu}_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} V_{\mu} + \bar{\theta}^2 \lambda_{\alpha} + 2 \theta_{\alpha} \left[\bar{\theta} \bar{\lambda} + \bar{\theta}^2 \left(D(y) - \frac{i}{2} \partial_{\mu} V^{\mu}(y) \right) \right] \\ + \underbrace{2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \theta^{\beta} \sigma^{\nu}_{\beta\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_{\mu} V_{\nu}}_{\textcircled{c}} + \underbrace{2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \theta^{\beta} \bar{\theta}^2 \partial_{\mu} \bar{\lambda}}_{\textcircled{d}}$$

$$\textcircled{c} 2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \theta^{\beta} \sigma^{\nu}_{\beta\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_{\mu} V_{\nu} \\ = -i \epsilon^{\dot{\alpha}\dot{\beta}} \bar{\theta}^2 \sigma^{\mu}_{\alpha\dot{\alpha}} \sigma^{\nu}_{\beta\dot{\beta}} \theta^{\beta} \partial_{\mu} V_{\nu} \\ = -i \epsilon^{\dot{\alpha}\dot{\beta}} \bar{\theta}^2 \epsilon^{\beta\gamma} \sigma^{\mu}_{\alpha\dot{\alpha}} \sigma^{\nu}_{\beta\dot{\beta}} \theta_{\gamma} \partial_{\mu} V_{\nu} \\ = i \bar{\theta}^2 \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\sigma}^{\nu\dot{\alpha}\gamma} \theta_{\gamma} \partial_{\mu} V_{\nu} \\ = i \bar{\theta}^2 (\sigma^{\mu} \bar{\sigma}^{\nu})_{\alpha}{}^{\gamma} \theta_{\gamma} \partial_{\mu} V_{\nu}$$

$$\textcircled{c} = i \bar{\theta}^2 \partial_{\mu} V_{\nu}(y) (\sigma^{\mu} \bar{\sigma}^{\nu})_{\alpha}{}^{\gamma} \theta_{\gamma}$$

$$\textcircled{d} 2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \bar{\theta} \partial_{\mu} \bar{\lambda} \theta^2 \\ = 2i \theta^2 \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \bar{\theta}^{\dot{\beta}} \partial_{\mu} \bar{\lambda}_{\dot{\beta}} \\ = i \theta^2 \bar{\theta}^2 \sigma^{\mu}_{\alpha\dot{\alpha}} \epsilon^{\dot{\alpha}\dot{\beta}} \partial_{\mu} \bar{\lambda}_{\dot{\beta}} = \underline{-i \theta^2 \bar{\theta}^2 \sigma^{\mu}_{\alpha\dot{\alpha}} \partial_{\mu} \bar{\lambda}^{\dot{\alpha}}}$$

$$D_{\alpha}^{(1)} V_{wz}^{(1)}(y, \theta, \bar{\theta}) = \sigma^{\mu}_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} V_{\mu}(y) + 2 \theta_{\alpha} \bar{\theta} \bar{\lambda}(y) + \bar{\theta}^2 \lambda_{\alpha}(y) \\ + \bar{\theta}^2 \left[2 \delta_{\alpha}^{\beta} D(y) + i (\sigma^{\mu} \bar{\sigma}^{\nu})_{\alpha}{}^{\beta} \partial_{\mu} V_{\nu}(y) - i \delta_{\alpha}^{\beta} \partial^{\mu} \partial_{\mu} V_{\nu} \right] \theta_{\beta} \\ - i \theta^2 \bar{\theta}^2 (\sigma^{\mu} \partial_{\mu} \bar{\lambda})_{\alpha}$$

* To make further progress we can rewrite this using

$$2(\sigma^{\mu\nu})_{\alpha}^{\beta} = i[-\delta_{\alpha}^{\beta}\eta^{\mu\nu} + (\sigma^{\mu}\bar{\sigma}^{\nu})_{\alpha}^{\beta}]$$

Proof: $\sigma^{\mu\nu} = \sigma^{\mu}\bar{\sigma}^{\nu} - \sigma^{\nu}\bar{\sigma}^{\mu}$

$$(\sigma^{\mu}\bar{\sigma}^{\nu})_{\alpha}^{\beta} + (\sigma^{\nu}\bar{\sigma}^{\mu})_{\alpha}^{\beta} = 2\eta^{\mu\nu}\delta_{\alpha}^{\beta}$$

$$- 2(\sigma^{\mu}\bar{\sigma}^{\nu})_{\alpha}^{\beta} - 2(\sigma^{\nu}\bar{\sigma}^{\mu})_{\alpha}^{\beta}$$

$$\Rightarrow [\sigma^{\mu}\bar{\sigma}^{\nu} - \sigma^{\nu}\bar{\sigma}^{\mu}]_{\alpha}^{\beta} = 2(\eta^{\mu\nu}\delta_{\alpha}^{\beta} - (\sigma^{\mu}\bar{\sigma}^{\nu})_{\alpha}^{\beta})$$

$$\text{but } \frac{i}{4}(\sigma^{\nu}\bar{\sigma}^{\mu} - \sigma^{\mu}\bar{\sigma}^{\nu})_{\alpha}^{\beta} = (\sigma^{\nu\mu})_{\alpha}^{\beta}$$

$$4i(\sigma^{\mu\nu})_{\alpha}^{\beta} = 2(\eta^{\mu\nu}\delta_{\alpha}^{\beta} - (\sigma^{\mu}\bar{\sigma}^{\nu})_{\alpha}^{\beta})$$

$$2(\sigma^{\mu\nu})_{\alpha}^{\beta} = i[(\sigma^{\mu}\bar{\sigma}^{\nu})_{\alpha}^{\beta} - \eta^{\mu\nu}\delta_{\alpha}^{\beta}]$$

Thus,

$$2\delta_{\alpha}^{\beta}D(y) + i[(\sigma^{\mu}\bar{\sigma}^{\nu})_{\alpha}^{\beta} - \delta_{\alpha}^{\beta}\eta^{\mu\nu}]\partial_{\mu}V_{\nu}$$

$$= 2\delta_{\alpha}^{\beta}D(y) + 2(\sigma^{\mu\nu})_{\alpha}^{\beta}\partial_{\mu}V_{\nu}(y)$$

$$= 2\delta_{\alpha}^{\beta}D(y) + (\sigma^{\mu\nu})_{\alpha}^{\beta}\partial_{\mu}V_{\nu} + (\sigma^{\nu\mu})_{\alpha}^{\beta}\partial_{\nu}V_{\mu}$$

$$= 2\delta_{\alpha}^{\beta}D(y) + (\sigma^{\mu\nu})_{\alpha}^{\beta}(\partial_{\mu}V_{\nu} - \partial_{\nu}V_{\mu})$$

Define $F_{\mu\nu} \equiv \partial_{\mu}V_{\nu} - \partial_{\nu}V_{\mu}$

$$\rightarrow = 2\delta_{\alpha}^{\beta}D(y) + (\sigma^{\mu\nu})_{\alpha}^{\beta}F_{\mu\nu}$$

$$D_{\alpha}^{(1)}V_{\omega\bar{\omega}}^{(1)}(y,\theta,\bar{\theta}) = \sigma^{\mu}_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}}V_{\mu} + 2\theta_{\alpha}\bar{\theta}^2 + \bar{\theta}^2\lambda_{\alpha}(y) - i\theta^2\bar{\theta}^2(\sigma^{\mu\nu}\partial_{\mu}\bar{\lambda}(y))_{\alpha}$$

$$+ \bar{\theta}^2(2\delta_{\alpha}^{\beta}D(y) + (\sigma^{\mu\nu})_{\alpha}^{\beta}F_{\mu\nu})\theta_{\beta}$$

With

$$D_{\dot{\alpha}}^{(1)} = -\partial_{\dot{\alpha}}$$

$$W_{\alpha} = -\frac{1}{4}\partial_{\dot{\alpha}}\partial^{\dot{\alpha}}(D_{\alpha}^{(1)}V_{\omega\bar{\omega}}^{(1)}(y,\theta,\bar{\theta}))$$

$$= -\frac{1}{4}\epsilon_{\dot{\alpha}\dot{\beta}}\partial^{\dot{\beta}}\partial^{\dot{\alpha}}(D_{\alpha}^{(1)}V_{\omega\bar{\omega}}^{(1)}(y,\theta,\bar{\theta})) = \frac{1}{4}\epsilon_{\dot{\alpha}\dot{\beta}}\partial^{\dot{\alpha}}\partial^{\dot{\beta}}(D_{\alpha}^{(1)}V_{\omega\bar{\omega}}^{(1)}(y,\theta,\bar{\theta}))$$

• First two terms vanish

$$\Rightarrow \partial^{\dot{\alpha}}\partial^{\dot{\beta}}(\bar{\theta}^2) = \partial^{\dot{\alpha}}\partial^{\dot{\beta}}(\bar{\theta}_{\dot{\gamma}}\bar{\theta}^{\dot{\gamma}}) = \epsilon^{\dot{\gamma}\dot{\delta}}\partial^{\dot{\alpha}}(\bar{\theta}_{\dot{\gamma}}\bar{\theta}_{\dot{\delta}})$$

$$\epsilon^{\dot{\gamma}\dot{\delta}}\partial^{\dot{\alpha}}\partial^{\dot{\beta}}(\bar{\theta}_{\dot{\gamma}}\theta_{\dot{\delta}}) = \epsilon^{\dot{\gamma}\dot{\delta}}\partial^{\dot{\alpha}}(\delta_{\dot{\gamma}}^{\dot{\beta}}\theta_{\dot{\delta}} - \bar{\theta}_{\dot{\gamma}}\delta_{\dot{\delta}}^{\dot{\beta}}) = \epsilon^{\dot{\gamma}\dot{\delta}}(\delta_{\dot{\gamma}}^{\dot{\beta}}\delta_{\dot{\delta}}^{\dot{\alpha}} - \delta_{\dot{\gamma}}^{\dot{\alpha}}\delta_{\dot{\delta}}^{\dot{\beta}})$$

$$= \epsilon^{\dot{\beta}\dot{\alpha}} - \epsilon^{\dot{\alpha}\dot{\beta}} = 2\epsilon^{\dot{\beta}\dot{\alpha}}$$

$$W_{\alpha} = \frac{1}{2}\epsilon^{\dot{\beta}\dot{\alpha}}\epsilon_{\dot{\alpha}\dot{\beta}}\left[\lambda_{\alpha}(y) + 2\theta_{\alpha}D(y) + \sigma^{\mu\nu}_{\alpha}{}^{\beta}\theta_{\beta}F_{\mu\nu} - i\theta^2\sigma^{\mu}_{\alpha\dot{\beta}}\partial_{\mu}\bar{\lambda}^{\dot{\beta}}(y)\right]$$

or

$$W_{\alpha} = \lambda_{\alpha}(y) + 2\theta_{\alpha}D(y) + (\sigma^{\mu\nu})_{\alpha}^{\beta}\theta_{\beta}F_{\mu\nu} - i\theta^2\sigma^{\mu}_{\alpha\dot{\beta}}\partial_{\mu}\bar{\lambda}^{\dot{\beta}}$$

and

$$W_{\dot{\alpha}} = \lambda_{\dot{\alpha}}(z) + 2\bar{\theta}_{\dot{\alpha}} D(z) - \varepsilon_{\dot{\alpha}\dot{\beta}} (\bar{\sigma}^{\mu\nu}\theta)^{\dot{\beta}} F_{\mu\nu}(z) + i\bar{\theta}^2 (\partial_{\mu}\lambda(z)\sigma^{\mu})_{\dot{\alpha}}$$

* Claim: $\bar{D}_{\dot{\alpha}} \bar{W}^{\dot{\alpha}} = D^{\alpha} W_{\alpha}$

proof: $\bar{D}_{\dot{\alpha}} \bar{W}^{\dot{\alpha}} = \varepsilon^{\dot{\alpha}\dot{\beta}} \bar{D}_{\dot{\alpha}} \bar{W}_{\dot{\beta}}$
 $= \varepsilon^{\dot{\alpha}\dot{\beta}} \bar{D}_{\dot{\alpha}} \left[-\frac{1}{4} D D \bar{D}_{\dot{\beta}} V(x, \theta, \bar{\theta}) \right]$
 $= \bar{D}_{\dot{\alpha}} \left[-\frac{1}{4} D D \bar{D}^{\dot{\alpha}} V(x, \theta, \bar{\theta}) \right]$
 $\rightarrow \bar{D}_{\dot{\alpha}} D^2 \bar{D}^{\dot{\alpha}} = D^{\alpha} \bar{D}^2 D_{\alpha}$
 $\rightarrow -\frac{1}{4} D^{\alpha} \bar{D}^2 D_{\alpha} V = D^{\alpha} W_{\alpha} \quad \checkmark$

* The Non-Abelian theories have:

$$\begin{aligned} W_{\alpha} &= -\frac{1}{4} \bar{D} \bar{D} (e^{-V} D_{\alpha} e^V) & W_{\alpha} &\mapsto e^{i\Lambda} W_{\alpha} e^{i\Lambda} \\ W_{\dot{\alpha}} &= \frac{1}{4} \bar{D} \bar{D} (e^V \bar{D}_{\dot{\alpha}} e^{-V}) & \bar{W}_{\dot{\alpha}} &\mapsto e^{i\bar{\Lambda}} \bar{W}_{\dot{\alpha}} e^{i\bar{\Lambda}} \end{aligned} \quad \Bigg| \text{Gauge}$$

$N=1$ Action:

• Chiral: $\mathcal{L}(\Phi^k)$

$$\mathcal{L} = \underbrace{K(\Phi^k, \Phi^{k\dagger})|_{\theta^2 \bar{\theta}^2}}_{\text{Kähler potential}} + \underbrace{\left(W(\Phi^k)|_{\theta^2} + W^{\dagger}(\Phi^{k\dagger})|_{\bar{\theta}^2} \right)}_{\text{Super-Potentials}}$$

only depends on Φ^k

$$K(\Phi^k, \Phi^{k\dagger})|_{\theta^2 \bar{\theta}^2} = \int d^4\theta K(\Phi^k, \Phi^{k\dagger}), \quad W(\Phi^k)|_{\theta^2} = \int d^2\theta W(\Phi^k)$$

* Thus

$$\int d^2\theta \Phi(x, \theta, \bar{\theta}) = F(x), \quad \int d^4\theta V(x, \theta, \bar{\theta}) = D(x)$$

Since

$$S = \int d^4\theta d^4x \mathcal{L}(\Phi)$$

We cannot have $\mathcal{L} = \Phi$ or $\mathcal{L} = \Phi + \Phi^{\dagger}$ and must have higher order terms.

• For a single Φ , a renormalisable theory is given by

$$K = \Phi^{\dagger} \Phi, \quad W = \frac{m}{2} \Phi^2 + \frac{g}{3} \Phi^3$$

(Wess-Zumino Model)

* Superfield Symmetries:

$$Q = P_{\mu} + i\gamma_{\mu} \partial_{\mu} + \dots$$

• $S = \int d^4x d^4\theta \Phi^\dagger \Phi$
 has a $U(1)$ symmetry $\Phi \mapsto e^{i\alpha} \Phi, \Phi^\dagger \mapsto e^{-i\alpha} \Phi^\dagger$

• Can also be local:

$$\Phi^i \mapsto \left(e^{i\alpha^a(x) T_a} \right)^i_k \Phi^k$$

Define $\Omega(x) = \alpha^a(x) T_a \leadsto$ Promote to a super field (how?)

$$\Phi \mapsto e^{i\Omega(x)} \Phi$$

$$\Phi^\dagger \mapsto \Phi^\dagger e^{-i\Omega^\dagger}$$

then

$$S \rightarrow \int d^4\theta d^4x \Phi^\dagger e^{i(\Omega - \Omega^\dagger)} \Phi$$

only valid if Ω is a vector super field.

$\leadsto$ Modify S :

Non-Abelian: Couple to V

$$S = \int d^4\theta d^4x \text{Tr}(\Phi^\dagger e^V \Phi e^{-V})$$

s.t.

$$e^V \mapsto e^{i\Omega^\dagger} e^V e^{-i\Omega}$$

* Super Yang Mills: $S = \frac{1}{4g_{YM}^2} \int d^4\theta d^4x \left[\text{Tr}(W^\alpha W_\alpha) \delta^2(\bar{\theta}) + \text{Tr}(\bar{W}_{\dot{\alpha}} \bar{W}^{\dot{\alpha}}) \delta^2(\theta) \right]$

• Introduce complex coupling constant $\tau \equiv \frac{2}{2a} + i \frac{4\pi}{g_{YM}^2}$

$$S_{S-M} = \frac{1}{8\pi^2} \int d^4x \text{Im} \left\{ \text{Tr} \left(\tau \int d^2\theta \text{Tr}(W^\alpha W_\alpha) \right) \right\}$$

• Note:

$$W^\alpha W_\alpha|_{\theta^2} = -2i\lambda(x) \sigma^\mu \partial_\mu \bar{\lambda}(x) - \frac{1}{2} F^{\mu\nu} F_{\mu\nu} + D^2 + \frac{i}{4} \epsilon_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$$

Renorm of the Wess Zumino model:

$$g_0 = g Z_g, \quad \Phi_0 = \Phi Z_\Phi$$

• WZM Action:

$$S = \int d^4x \left\{ i \partial_\mu \bar{\Phi}_i(x) \bar{\sigma}^\mu \Phi_i(x) - \Phi_i^* \square \Phi_i - \frac{1}{2} W_{ij} \Phi_i(x) \Phi_j(x) - \frac{1}{2} \bar{W}_{\dot{i}\dot{j}} \bar{\Phi}_{\dot{i}} \bar{\Phi}_{\dot{j}} + |W_i|^2 \right\}$$

$$|W_i|^2 = |g + m\phi(x) + \lambda\phi^2|^2$$

• $Z_g Z^3 = 1$ (shown order by order)

and S_0

$$(g/3) \mathbb{F}^3 \rightarrow g_0 Z^3/3 \mathbb{F}^3 \rightarrow [(g Z_g Z^3)/3] \mathbb{F}^3$$

$$= g \mathbb{F}^3$$

$$\beta(g) \equiv \frac{\partial g}{\partial \ln \mu}, \quad \gamma(g) \equiv \frac{1}{2} \frac{\partial \ln(Z_g)}{\partial \ln \mu}$$

$$\rightarrow \beta(g) - 3\gamma(g)g = 0$$

NSVZ β -fn

$$\beta(g) = -\frac{g^3}{8\pi^2} \left(\frac{3C(\text{adj}(G)) - \sum_A C(R_A)(1-2\gamma_A)}{1 - g^2 C(\text{adj}(G))/(8\pi^2)} \right)$$

γ_A : Anomalous dimension of $\mathbb{F}^A$ ($\gamma_A \equiv \frac{1}{2} \frac{\partial Z_{\mathbb{F}^A}}{\partial \ln \mu}$)

R_A : Rep of $\mathbb{F}^A$ (i.e. Spin , $\underline{1}$, vect , adj , ...)

Maximally super symmetric Yang Mills in $d=4$:

- For $d=4$, $s_{\text{max}} \leq 1$, $N=4$ is the maximal amount of susy possible

$$s_{\text{max}} = \lambda_0 + \frac{N}{2}$$

$$\lambda_0 \rightarrow \lambda_0 + \frac{1}{2} \rightarrow \lambda_0 + \frac{3}{2} \rightarrow \dots \rightarrow \lambda_0 + \frac{N}{2}$$

$$s_{\text{max}} = 1: \quad 1 = \lambda_0 + 2 \Rightarrow \lambda_0 = -1$$

$|p, -1\rangle$ (massless lowest weight state)

Massless: 2^N : $2^4 = 16$ fermionic supercharges

Q: Why no CPT? Appears the lowest weight up to $d=1$ is already fully susy invariant/symmetric so not required?

- $Q_a^i, \bar{Q}_{\dot{a}i}$, all massless states with helicities between -1 and 1 are generated by acting w/ no more than $N_{\text{max}} = 4$ different supercharges
 $\Rightarrow$ Theory is constructed from only Vector Multiplets / Field strengths and has no central charges.

$N=4$ Gauge supermultiplet: $(A_\mu, (4) \lambda_a^\alpha(x), (6) \phi^i)$

(vectors) (scalars)

$$\underline{2} \oplus 4 \cdot \underline{2} \oplus 6 \cdot \underline{1}$$

$\bullet$ $8B \oplus 8F$ D.O.F. $a=1, \dots, 4$

$SO(2) \cong U(1)$ Massless reps $i=1, \dots, 6$

$\#$ Theory also possesses a R-symmetry group $SU(4)_R$

Vector	A_μ	$\underline{1}$ (singlet)	$\bullet$	$SU(4)_R$
Weyl spinor	$\lambda_a^\alpha \quad a=1, \dots, 4$	$\underline{4}$ (fundamental)	$\square$	
	ϕ^i	$\underline{6}$ (antisymmetric)	$\overline{\square}$	

$N=4$ Super-Yang Mills in $N=1$ superspace:

• Requires:

(a) 3 chiral superfields Φ^i , $i=1,2,3$

(b) 1 Gauge superfield V w/ Gauge field W_α .

$$\leadsto \text{Tr}(T_a T_b) = \delta_{ab}$$

$$S_{N=4} = \int d^4x \text{Tr} \left[\int d^4\theta \Phi^i e^V \Phi^i e^{-V} + \frac{1}{8\pi} \text{Im}(\tau) \int d^2\theta W^\alpha W_\alpha \right. \\ \left. + i g_{YM} \frac{\sqrt{2}}{3!} \int d^2\theta \epsilon_{ijk} \Phi^i [\Phi^j, \Phi^k] + \text{h.c.} \right]$$

① $\Phi^i e^V \Phi^i e^{-V}$, $V = V^a T_a$

• Φ^i are in the adjoint rep of the theory N^2-1 

• Let's use Conventions from other texts:

$$\text{Tr}[\Phi^i e^V \Phi^i]_D$$

$\leadsto$ Go to the WZ Gauge:

$$(V^a)_{WZ} = -\theta \sigma^m \bar{\theta} A_m^a (1 - i \bar{\theta}^2 \theta \lambda^a + i \theta^2 \bar{\theta} \bar{\lambda}^a + \frac{1}{2} \theta^2 \bar{\theta}^2 D^a)$$

$$e^V = 1 + V + \frac{1}{2} V^2$$

$$V_{WZ}^2 = -\frac{1}{2} \theta^2 \bar{\theta}^2 A_m A^m$$

$$\Phi^i = \phi^i(x) + i \theta \sigma^m \bar{\theta} \partial_m \phi^i + \frac{1}{4} \theta^2 \bar{\theta}^2 \partial_\rho \partial^\rho \phi^i + \sqrt{2} \theta \psi^i + \frac{1}{2} \theta^2 \bar{\theta}^2 \partial_\mu \psi^i \bar{\theta} + \theta^2 F^i$$

$\leadsto$ Find D term

• I believe we take ϕ to be in the adjoint rep $\phi^i = (\phi^i)^a T_a$

$$\phi_j = \phi^a (T_a)^j_i \mapsto (\phi')^j_i = (e^{i\psi^b T_b})^j_k \phi^a (T_a)^k_l (e^{-i\psi^c T_c})^l_i$$

$$\Phi \mapsto U(x) \Phi(x) U^\dagger(x)$$

$$D_\mu \phi = \partial_\mu \phi + i [A_\mu, \phi]$$

$$\text{Tr}(W^\alpha W_\alpha) \Big|_F = D^a D^a - 2 i \lambda^a \sigma^m D_\mu \bar{\lambda}^a - \frac{1}{2} F_{\mu\nu}^a F_{\mu\nu}^a + \frac{i}{2} F_{\mu\nu}^a \tilde{F}^{\mu\nu}_a$$

$$\text{and } \bar{D} = \frac{\partial}{\partial \bar{\theta}} + i \frac{u\pi}{g^2_{YM}}$$

$$\Rightarrow -\frac{1}{8\pi} \text{Im} \left\{ \int d^2\theta W^\alpha W_\alpha \right\} \Rightarrow \frac{1}{8\pi} \left(\frac{\partial}{\partial \bar{\theta}} + i \frac{u\pi}{g^2_{YM}} \right) \left(D^a D^a - 2 i \lambda^a \sigma^m D_\mu \bar{\lambda}^a - \dots \right) \\ = \left(\frac{1}{8\pi} \right) \left[\left(\frac{\partial}{\partial \bar{\theta}} \right) \left(-2 i \lambda^a \sigma^m D_\mu \bar{\lambda}^a + \frac{i}{2} F_{\mu\nu}^a \tilde{F}^{\mu\nu}_a \right) + \left(\frac{i u \pi}{g^2_{YM}} \right) \left(D D - \frac{1}{2} F_{\mu\nu}^a F^{\mu\nu}_a \right) \right]$$

$$\therefore \frac{1}{8\pi} \text{Im} \{ D^2 W^\alpha W_\alpha \} = \frac{1}{8\pi} \left[\left(\frac{\partial}{\partial \bar{\theta}} \right) \left(-2 \lambda^a \sigma^m D_\mu \bar{\lambda}^a + \frac{1}{2} F_{\mu\nu}^a F^{\mu\nu}_a \right) + \frac{u\pi}{g^2_{YM}} \left(-\frac{1}{2} F_{\mu\nu}^a \tilde{F}^{\mu\nu}_a + D D \right) \right]$$

off by factor of 2:

$$\mathcal{L} = \left(\frac{1}{8\pi^2} \right) \left(\frac{1}{2} F_{\mu\nu} F^{\mu\nu} - 2 \lambda \sigma^{\mu\nu} D_{\mu} \bar{\lambda} \right) + \left(\frac{2}{g_{YM}^2} \right) \left(D\phi - \frac{1}{2} F_{\mu\nu} \tilde{F}^{\mu\nu} \right)$$

• Note $\lambda \sigma^{\mu\nu} D_{\mu} \bar{\lambda}$ and $g_{YM}^{-2} D^2$ terms are "missing" from the expansion

$N=4$:

$$\mathcal{L} = \text{Tr} \left[-\frac{1}{2g_{YM}^2} F_{\mu\nu} F^{\mu\nu} + \frac{1}{16\pi^2} F_{\mu\nu} \tilde{F}^{\mu\nu} - i \bar{\lambda}^a \sigma^{\mu\nu} D_{\mu} \lambda^a - \sum_i D_{\mu} \phi^i D^{\mu} \phi^i + g_{YM} \sum_{a,b,i} C_{ab,i}^{ab} \lambda_a [\phi^i, \lambda_b] \right. \\ \left. + g_{YM} \sum_{a,b,i} \bar{C}_{iab} \bar{\lambda}^a [\phi^i, \lambda^b] + \frac{g_{YM}^2}{2} \sum_{ij} [\phi^i, \phi^j]^2 \right]$$

$$F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu} + i [A_{\mu}, A_{\nu}], \quad D_{\mu} \cdot = \partial_{\mu} \cdot + i [A_{\mu}, \cdot], \quad \tilde{F}_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}$$

• $C_{ab,i}^{ab}$: Clebsch-Gordan coefficients that couple two 4 reps of $SU(4)_R$

↳ to one 6 $\square$

↳ 6-dim generalizations of $\sigma^{\mu\nu}$

* Invariant under susy:

$$\delta_{\epsilon} \phi^i = [\epsilon_a^{\alpha} Q_{\alpha}^a, \phi^i] = \epsilon_a^{\alpha} C^{iab} \lambda_{\alpha b}$$

$$\delta_{\epsilon} \lambda_{ab} = [\epsilon_a^{\alpha} Q_{\alpha}^a, \lambda_{ab}] = F_{\mu\nu}^+ \epsilon_{ab} (\sigma^{\mu\nu})_{\beta}^{\alpha} + [\phi^i, \phi^j] \epsilon_{\beta a} (C_{ij})^{\alpha}_{\beta}$$

$$\delta_{\epsilon} \bar{\lambda}_{\dot{a}}^b = [\epsilon_a^{\alpha} Q_{\alpha}^a, \bar{\lambda}_{\dot{a}}^b] = C_i^{ab} \epsilon_a^{\alpha} \sigma_{\alpha\dot{\beta}}^{\mu} D_{\mu} \phi^i$$

$$\delta_{\epsilon} A_{\mu} = [\epsilon_a^{\alpha} Q_{\alpha}^a, A_{\mu}] = \epsilon_a^{\alpha} (\sigma_{\mu})_{\alpha}^{\beta} \bar{\lambda}_{\dot{\beta}}^a \quad SO(6) \cong SU(4)$$

↪ $F_{\mu\nu}^+$: self dual part of $\frac{1}{2} (F_{\mu\nu} + \tilde{F}_{\mu\nu})$

↪ $(C_{ij})^a_b$: related to bilinears in Clifford Dirac mat of $SO(6)_R$

$N=4$ SYM From dim reduction:

$N=1, D=10, SYM$:

$$S_{10D} = \int d^{10}x \text{Tr} \left(-\frac{1}{2} F_{mn} F^{mn} + \frac{i}{2} \bar{\Psi} \Gamma^m D_m \Psi \right)$$

$$F_{mn} = \partial_m A_n - \partial_n A_m + ig [A_m, A_n]$$

Ψ : Majorana-Weyl,

$SO(8)$

2^4 : 16

Ψ : 16-real D.O.F

• Both F_{mn} and Ψ are in the adjoint and $D_m \Psi = \partial_m \Psi + ig [A_m, \Psi]$

↪ Above action is invariant under

$$\delta_{\epsilon} A_m^i = i \bar{\epsilon} \Gamma_m^i \Psi^i$$

$$\delta_{\epsilon} \lambda^i = \Gamma_{mn} (F^i)^{mn} \epsilon$$

* Can dimensionally reduce this theory from 10D to 6-d on T^4 by keeping only

the massless modes: preserves the # of susy

$$A_{\hat{\mu}} = \begin{cases} A_m(x), & \hat{\mu} = m = 0, \dots, d-1 \\ \phi_i(x), & \hat{\mu} = i = d, \dots, 9 \end{cases}$$

$$F_{\hat{m}\hat{n}} \rightarrow F_{\hat{m}\hat{n}} F^{\hat{m}\hat{n}} = F_{m\hat{n}} F^{m\hat{n}} + F_{i\hat{n}} F^{i\hat{n}} = F_{mn} F^{mn} + F_{mj} F^{mj} + F_{im} F^{im} + F_{ij} F^{ij}$$

$$= F_{mn} F^{mn} + 2 F_{mj} F^{mj} + F_{ij} F^{ij}$$

$$F_{\hat{m}\hat{n}} = \partial_{\hat{m}} A_{\hat{n}} - \partial_{\hat{n}} A_{\hat{m}} + ig [A_{\hat{m}}, A_{\hat{n}}]$$

$F_{mn} \vee$

$$F_{im} = \partial_i A_m - \partial_m \phi^i + ig [\phi^i, A_m]$$

$$F_{ij} = \partial_i \phi_j - \partial_j \phi_i + ig [\phi_i, \phi_j]$$

Thus:

$$F_{mn} = \partial_m A_n - \partial_n A_m + ig [A_m, A_n]$$

$$F_{mi} = \partial_m \phi^i + ig [A_m, \phi^i] \equiv D_m \phi^i$$

$$F_{ij} = ig [\phi_i, \phi_j]$$

$$\bullet F_{mi} F^{mi} = D_m \phi^i D^m \phi_i$$

$$F_{ij} F^{ij} = -g^2 [\phi_i, \phi_j]^2$$

$$\Rightarrow -\frac{1}{2} \text{Tr} (F_{\hat{m}\hat{n}} F^{\hat{m}\hat{n}}) = -\frac{1}{2} \text{Tr} (F_{mn} F^{mn} + 2 D_m \phi^i D^m \phi_i - g^2 [\phi_i, \phi_j]^2)$$

$$= \text{Tr} \left(-\frac{1}{2} F_{mn} F^{mn} - D_m \phi^i D^m \phi_i + \frac{g^2}{2} [\phi_i, \phi_j]^2 \right) \quad *$$

$$\Gamma^{\hat{m}} = \gamma^m \otimes \hat{\gamma}^i \in \text{Spin}(4) \otimes \text{Spin}(6)$$

$M^4 \otimes T^6$

$$\psi \frac{i}{2} \bar{\psi} \Gamma^{\hat{m}} D_{\hat{m}} \psi = \frac{i}{2} \bar{\psi} \Gamma^m D_m \psi + \frac{i}{2} \bar{\psi} \Gamma^i D_i \psi \quad \psi = \begin{pmatrix} \chi_\alpha \\ \bar{\lambda}_{\dot{\alpha}} \end{pmatrix}$$

$$\left(D_i \psi = \partial_i \psi + ig [\phi_i, \psi] = ig [\phi_i, \psi] \right)$$

$$\Rightarrow \frac{i}{2} \bar{\psi} \Gamma^m D_m \psi + \frac{i}{2} \bar{\psi} \Gamma^i ig [\phi_i, \psi]$$

$$= \frac{i}{2} [\bar{\psi} \Gamma^m (\partial_m \psi + ig [A_m, \psi]) + \bar{\psi} \Gamma^i ig [\phi_i, \psi]]$$

$$\sum_2^M = \sigma^m$$

10D $\Gamma^{\hat{m}}$: $\Gamma^{\hat{m}} = \tilde{V}_2^{\mu} \otimes E_{\mu}$

$$\Gamma^{7+i} = 1 \otimes E_i$$

$$\mu = 0, 1$$

$$i = 1, \dots, 8$$

$$E_{\mu} = \sigma_2^{\otimes 4}$$

$$(\lambda, \bar{\lambda}) \begin{pmatrix} 0 & \sigma^m \\ \bar{\sigma}^m & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ \bar{\lambda} \end{pmatrix} = (\lambda, \bar{\lambda}) \begin{pmatrix} \sigma^m \bar{\lambda} \\ \bar{\sigma}^m \lambda \end{pmatrix} = \lambda \sigma^m D_m \bar{\lambda} - \bar{\lambda} \bar{\sigma}^m D_m \lambda$$

$$= -2 \bar{\lambda} \bar{\sigma}^m D_m \lambda$$

$$\rightarrow -i \bar{\lambda} \sigma^{\mu} D_{\mu} \lambda + \bar{\psi} \Gamma^i i g [\phi_i, \psi]$$

$$\textcircled{b} \quad (\lambda, \bar{\lambda}) \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \lambda \\ \bar{\lambda} \end{pmatrix} ([\phi_i, \tilde{\tau}])$$

$$\Gamma^i = \begin{pmatrix} -2C^{ab}_i & 0 \\ 0 & -2\bar{C}_{iab} \end{pmatrix}$$

$$(\lambda_a, \bar{\lambda}^a) \begin{pmatrix} -2C^{ab}_i & 0 \\ 0 & -2\bar{C}_{iab} \end{pmatrix} \begin{pmatrix} \lambda_b \\ \bar{\lambda}^b \end{pmatrix} ([\phi_i, \tilde{\tau}])$$

$$= (\lambda_a, \bar{\lambda}^a) \begin{pmatrix} -2C^{ab}_i & 0 \\ 0 & -2\bar{C}_{iab} \end{pmatrix} \begin{pmatrix} \lambda_b \\ \bar{\lambda}^b \end{pmatrix} ([\phi_i, \tilde{\tau}])$$

$$= -2 (\lambda_a C^{ab}_i \lambda_b + \bar{\lambda}^a \bar{C}_{iab} \bar{\lambda}^b) [\phi_i, \tilde{\tau}]$$

$$\therefore -\frac{g}{2} \bar{\psi} \Gamma^i [\phi_i, \psi] = -\frac{g}{2} [-2 \lambda_a C^{ab}_i [\phi_i, \lambda_b] - 2 \bar{\lambda}^a \bar{C}_{iab} [\phi_i, \bar{\lambda}^b]]$$

$$= g_{YM} \sum_{a,b,i} \lambda_a C^{ab}_i [\phi_i, \lambda_b] + g_{YM} \sum_{a,b,i} \bar{\lambda}^a \bar{C}_{iab} [\phi_i, \bar{\lambda}^b]$$

Note: I have no basis for $\Gamma^i = -2 \begin{pmatrix} C^{ab}_i & 0 \\ 0 & \bar{C}_{iab} \end{pmatrix}$ I've really just reverse engineered the expression based on the form of $\mathcal{L}_{D=4, N=4}$.

Properties of N=4 SYM:

- Coupling g_{YM} : dimensionless, S_4 is scale invariant at the classical level i.e. its weyl / conformal invariant. $[A_{\mu}] = 1, [\lambda] = 3/2, [\phi_i] = 1$

- Scale invariant after quantisation: UV-finite and β -fn vanishes to all orders in pert. theory. Theory is conformally invariant under $SO(4,2)$

$$G = G_0 \times SU(4)_R \times SO(4,2) \times SUSY$$

- Invariant under S-duality Group $SL(2, \mathbb{Z})$: Moduli space of T^2

ω_4 acts on τ :

$$\tau \mapsto \frac{a\tau + b}{c\tau + d}, \quad ad - bc = 1, \quad a, b, c, d \in \mathbb{Z}$$

$$PSL(2, \mathbb{Z})$$

$\leadsto$ Strong-weak duality

ex: $\tau = \frac{4\pi i}{g_{YM}^2}$ (since we gauge $\mathcal{L} = 0$) $\{a, b\}$

[cd]

Let $b = -c = 1, \quad a = d = 0 : \quad M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \epsilon_{\alpha\beta}$

"metric" on $\text{PSL}(2, \mathbb{Z})$

$$\tau \mapsto \frac{1}{-\tau} = \frac{ig_{\text{YM}}^2}{u\bar{v}} \Rightarrow g_{\text{YM}} \mapsto \frac{u\bar{v}}{g_{\text{YM}}}$$

• $N=4$ has two classes of vac: Scalar potential must vanish in the susy ground state $[\phi_i, \phi_i]^2 \geq 0$

$$[\phi_i, \phi_i] = 0 \Leftrightarrow \phi_i = \text{const.} \quad \forall i, j \in \{1, \dots, 6\}$$

i) Superconformal phase: $\langle \phi_i \rangle = 0$

ii) Coulomb phase: $\exists ! \phi_{\tilde{v}}$ s.t. $\langle \phi_{\tilde{v}} \rangle \neq 0$

$\Rightarrow$ Conf. inv. is broken since $\langle \phi_{\tilde{v}} \rangle$, a length scale, is introduced (sup-conf-breaking)

$$SU(N) \rightarrow U(1)^{N-1}$$

Super Conformal Symmetry: